Spaceflight brain: study workflow

Biological question: do myeloid RNA changes reflect cell mixture, transcriptional remodeling, or sample handling?

A Verified data and biological replicates

Original project

OSD-561 / 562: cerebellum + hippocampus
53 libraries · 29 mice · 24 mice with paired regions

OSD-682 / 685 / 698 / 699: GeoMx
4 regions · 48 ROIs · animal links unresolved

Mission comparisons

RR8: 47 libraries · 8 mice · 122,084 cells (GEO gate)
RR10: 10 paired mice · RR9: 12 mice

Added: RR23 cerebellum · 26 mice
Added: MHU-3 cerebrum · 24 mice (WT / Nrf2KO)
Added: RRRM-2 cerebral region · 18 different mice

RR3 multimodal context

OSD-352: 21,178 nuclei · 5 mice
RNA + ATAC measured from the same nuclei

Visium: 29,839 spots · 6 mice
12 technical sections · verified coordinates

151 corrected human targets → 143 primary mouse symbols | ISA identities · count QC · technical-library collapse

B Five biological and analytical evidence branches

QUESTION	DATA AND ANALYSIS	CURRENT EVIDENCE
1 Biological specificity Which sensor, adaptor and execution transcripts change?	RR8: six enriched populations · whole-genome count models edgeR QL + limma/voom · GEO, article and alternative QC All 151 targets × populations × age contrasts retained	Forebrain candidate RNA program: AIM2 higher; PYCARD, IL18 and NINJ1 lower.
2 State or cell mixture? Does the signal persist in comparable expression states?	Non-panel reference mixtures: held-donor NNLS / ridge New: balanced, exposure-blind PCA states (K=4 / K=6) Mouse-state pseudobulk · cell coverage · fixed-composition checks	PYCARD persists in primary shared states. Technical / finer-state sensitivity limits the claim.
3 Transfer across missions Does the association transfer across region, strain and control?	RR3 / RR10 / RR9 context + RR23 / MHU-3 / cerebral counts Habitat and vivarium contrasts kept separate Nrf2 genotype interaction · 50 complete mouse-deletion fits	RR23 AIM2: flight > habitat, deletion-stable. PYCARD corrected support varies by control / family.
4 Spatial and regulatory support Is there regional localization and promoter agreement?	Original GeoMx contrasts + RR3 donor-linked Visium Spatial neighbours separated from exposure tests Donor RNA and promoter ATAC · same-mission dependence	Local spatial structure is present. Corrected spatial / promoter flight effects unconfirmed.
5 Inferential robustness Which results survive method and assumption changes?	Three bulk engines · complete planned testing families Mouse deletion · HC3 intervals · exact animal permutations 50 Hallmark sets · correlation and residual-df sensitivity	Report effects, uncertainty and failed checks. RNA associations remain observational.

C Next mechanistic test · proposed laboratory validation

Matched handling + independently identified myeloid cells → multiplex RNA / protein → ASC, caspase and cytokine readouts → perturbation / rescue

Aim: test sensor–adaptor divergence. Computational evidence does not establish inflammasome activity or a causal protective response.

Limits carried through every branch: terminal bulk preservation; RR8 QC threshold / dissection day; RR23 collection dates; small mouse cohorts and unresolved GeoMx animal identities.